Fingerprint-Based Blood Group Detection Using Deep Learning

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Abstract—Particularly in emergency situations where prompt and precise transfusion decisions are needed, blood group identification is a therapeutically crucial process. Despite being dependable, conventional serological techniques are intrusive, time-consuming, and require specific laboratory equipment. Convolutional Neural Networks (CNNs) are used in this paper to develop a non-invasive fingerprint-based blood group detection method. Prior to minutiae-based feature extraction, fingerprint pictures are preprocessed using ridge segmentation, histogram equalization, and binarization. Multiple CNN architectures—ResNet, VGG16, AlexNet, and LeNet—are evaluated on a labeled dataset of eight blood group classes (A+, A−, B+, B−, AB+, AB−, O+, O−). ResNet achieves a training accuracy of 95.54% and validation accuracy of 81.42% at epoch 20, while VGG16 reaches 98.0% training accuracy. The system demonstrates potential practical applicability by real-time forecasts that show confidence scores higher than 99.8%. The suggested method offers a quick, affordable, and scalable substitute for blood group determination in emergency medicine, forensics, and healthcare.

Keywords—blood group detection, fingerprint biometrics, convolutional neural network, deep learning, image processing, feature extraction, non-invasive diagnosis.

I. INTRODUCTION

Important medical operations, such as organ transplants, blood transfusions, and emergency care, depend on blood group identification. Eight clinically relevant blood types are defined by the ABO and Rh systems. Life-threatening hemolytic

reactions can result from mismatched transfusions, which emphasizes the necessity for quick, accurate, and non-invasive identification techniques.

Even though they are accurate, traditional serological agglutination tests require specialized laboratory equipment, trained personnel, and needle-based blood sampling. These limitations make them unfeasible in emergency or resource-constrained situations. An appealing substitute is provided by the increasing use of computer vision and artificial intelligence in medical diagnosis. Fingerprints are unique and permanent biometric characteristics that don't change over the course of a person's life.

According to new research, their ridge patterns—loops, whorls, and arches—encode information that may be correlated with blood group characteristics. Sweat released by fingerprint ridges is thought to contain blood group system proteins and antigens [1].

A CNN-based fingerprint blood group recognition system trained on eight blood group classes is presented in this study. We provide performance across 20 training epochs for four architectures: ResNet, VGG16, AlexNet, and LeNet. With encouraging validation results and over 95% training accuracy, the system shows promise for practical application in emergency medical care, forensic science, and remote diagnostics.

II. LITERATURE REVIEW

Statistical techniques were used in the early research on the link between blood group and fingerprint. Using systematic random sampling, Elshafie et al. discovered that loops were the most common pattern (62.84%), possibly correlated with gender and blood group [2]. Using ink-based techniques, Khalifa et al. found that A-type individuals had a higher whorl frequency, whereas O-type persons primarily display loop patterns [3]. On 138 samples, Ramrekh et al. did not find any statistically significant correlation, underscoring the difficulty of finding a universal relationship.

According to Shivhare et al.'s dermatoglyphic research, loop patterns are most prevalent among Rh-positive people.

These studies were enlightening, but they were not scalable and relied on manual classification. Partial automation was introduced using image processing techniques: Pimenta et al. created a spectrophotometric device that produced results in less than five minutes, while Dalvi et al. created an automated blood typing system based on agglutination, both of which still required physical blood samples [4]. A major change was brought about by machine learning techniques. For fingerprint-based blood group prediction, D. Siva et al. used GLCM features with a neural network classifier, outperforming statistical baselines.

The introduction of deep learning further accelerated progress in this field. Saponara et al. used a CNN autoencoder to recreate fingerprints with 96.5% accuracy [5]. The feasibility of deep learning for this classification job was demonstrated by Nihar et al.'s proposal of a Gabor filter-based feature extraction pipeline feeding into CNN architectures [1].

The lack of a widely recognized biological mechanism connecting fingerprint morphology to blood type, inconsistent preprocessing pipelines across studies, sensitivity to fingerprint image quality, and the scarcity of publicly accessible labeled datasets combining fingerprints with blood group annotations are some of the enduring challenges. The current approach uses a common preprocessing pipeline and transfer-learning-compatible structures to fill these gaps.

TABLE I. RELATED WORKS SUMMARY

Author	Method	Findings	Limitations
Patil et al. (2021)	ML / Linear Regression	62% accuracy; correlation between fingerprints and blood group	Limited dataset; low accuracy
Alshehri et al. (2018)	Gabor-HOG descriptors	Sensor-independent feature extraction	Degrades with distorted images
Saponara et al. (2021)	CNN Autoencoder	96.5% accuracy on reconstructio n	No classification
Nihar et al. (2024)	Gabor + CNN	Ridge frequency + CNN pipeline feasible	Requires larger dataset

III. PROPOSED METHODOLOGY

A. System Overview

Image acquisition, preprocessing, feature extraction, data augmentation, CNN-based classification, and blood group prediction comprise the six steps of the suggested system's workflow. In order to meet the input requirements of pretrained architectures, fingerprint images are standardized to 224 by 224 pixels. Eight classes are targeted by the system: A+, A−, B+, B−, AB+, AB−, O+, and O−.

B. Data Collection and Augmentation

Owing to the lack of publicly accessible labeled fingerprint–blood group databases, information was gathered from carefully selected fingerprint collections and enhanced by a physical survey. For training and testing, the dataset was divided 80:20. Augmentation techniques such as rotation, scaling, horizontal flipping, and Gaussian noise injection were used to successfully multiply

dataset variety in order to combat limited data volume and prevent overfitting.

C. Preprocessing

Three preprocessing steps are applied to raw fingerprint pictures. Ridge Segmentation uses variance-based thresholding to distinguish foreground ridges from background areas. In order to improve contrast and highlight subtle ridge details, Histogram Equalization redistributes pixel intensity levels. Binarization creates a high-contrast black-and-white fingerprint that makes downstream feature extraction easier by converting the resulting grayscale image to a binary form.

D. Feature Extraction

Feature extraction uses thinning to reduce ridge width to a single pixel while preserving the topological structure of fingerprint ridges. Although minutiae features such as ridge endings and bifurcations were analyzed during preprocessing, the CNN models were trained directly on preprocessed fingerprint images rather than handcrafted feature vectors. The preprocessing steps enhance ridge patterns and reduce noise, enabling the networks to automatically learn discriminative features for blood group classification.

E. CNN Architectures

Four architectures are assessed. LeNet-5 uses average pooling and tanh activations over seven layers to give a lightweight baseline appropriate for small datasets. AlexNet, which has three completely connected and five convolutional layers, introduces dropout regularization and ReLU activations. VGG16 uses uniform 3x3 convolutions and 16 weight layers to extract deep spatial features. In order to handle vanishing gradients and enable reliable training of deeper networks, ResNet uses skip connections. Every model is trained for up to 20 epochs with a 224x224 input resolution using the Adam optimizer, categorical cross-entropy loss, and a batch size of 32.

TABLE II. CNN MODEL HYPERPARAMETERS

Parameter	Value
Input Size	224 × 224
Epochs	20
Batch Size	32
Optimizer	Adam
Loss Function	Categorical Cross-Entropy
Train/Test Split	80% / 20%

IV. IMPLEMENTATION AND RESULTS

A. Development Environment

The system was implemented in Python 3.x using TensorFlow/Keras for model construction and training, OpenCV for image preprocessing, and NumPy/Pandas for data management. Matplotlib was used for training curve visualization. All experiments were conducted on a GPU-enabled workstation to accelerate convolutional operations.

B. Training Performance

Epochs 1, 2, 5, 10, 15, and 20 were used to track training accuracy and validation accuracy. Stable convergence was confirmed by all architectures' monotonically rising accuracy and falling loss throughout training. However, AlexNet's consistently poor performance (~16.4% training accuracy at epoch 20) indicates that, absent major architectural changes, its design is unsuitable for this classification issue. The performance of the three top-performing architectures at significant epochs is summarized in Table III.

TABLE III. MODEL PERFORMANCE ACROSS EPOCHS

Epoch	Model	Train Acc	Val Acc	Train Loss	Val Loss
1	ResNet	0.4125	0.6492	1.5054	0.8991
1	VGG16	0.4765	0.6300	1.4881	0.9685
5	ResNet	0.7957	0.7667	0.5272	0.5778
5	VGG16	0.8050	0.7442	0.5160	0.6732
10	ResNet	0.8957	0.7483	0.2849	0.6633
10	VGG16	0.8948	0.7283	0.2894	0.7657
15	ResNet	0.9115	0.8050	0.2295	0.5474
15	VGG16	0.9427	0.7417	0.1588	0.9221
20	ResNet	0.9554	0.8142	0.1341	0.5838
20	VGG16	0.9800	0.7375	0.0714	1.0450

C. Checkpoint Analysis

With validation accuracy plateauing at 81.42%, ResNet checkpoint curves provide a smooth, monotonic accuracy improvement from 41.25% at epoch 1 to 95.54% at epoch 20. Training loss consistently reduces while validation loss

stabilizes, according to the related loss curves, suggesting that the model generalizes without experiencing significant overfitting. VGG16 has the best training accuracy (98.0%), but after epoch 15, its validation loss increases, indicating overfitting caused by its large number of parameters and little dataset.
LeNet initially shows rapid accuracy growth (86.48% at epoch 5) but validation accuracy remains below 50%, and validation loss diverges to 4.85 by epoch 20—a clear indicator of overfitting on a dataset insufficient for its fully connected layers. These findings affirm that ResNet, with residual connections enabling gradient flow, provides the best generalization among evaluated architectures.

D. Prediction Results

The viability of the technique is confirmed by live predictions. Blood group classifications for test fingerprints were A+ with 99.83% confidence, A+ with 99.86% confidence, and AB− with 83.10% confidence. The model's taught discriminative capacity is confirmed by these high-confidence predictions on unseen fingerprint photos. The known scarcity of this blood group in the training distribution and the resulting class imbalance difficulty are consistent with the relatively lower confidence for AB−.

V. DISCUSSION

The experimental findings verify that deep learning architectures, specifically ResNet, are capable of extracting significant discriminative patterns from fingerprint photos in order to classify blood groups. ResNet's 95.54% training accuracy and 81.42% validation accuracy demonstrate a competent model functioning within dataset limitations. The discrepancy between training and validation accuracy across all architectures indicates that the main barrier is not inadequate architecture but rather a lack of diverse datasets.
The findings are consistent with related research: Gabor filter preprocessing with LeNet/AlexNet designs was shown to be similarly feasible by Nihar et al. [1]. By methodically testing four architectures, our comparative study adds value and shows that residual learning is essential for consistent convergence on this classification task. Given its known sensitivity to data scale and lack of batch or residual normalization components, AlexNet's inability to learn (≈16.5% accuracy over all epochs) makes sense.
The suggested method's non-invasiveness offers observable benefits over conventional serological techniques, including the ability to interface with current fingerprint sensors or smartphone cameras, produce results in a matter of seconds, and require no blood collection. Emergency room triage, military field medicine, catastrophe victim identification, and point-of-care diagnostics in rural healthcare facilities are among possible deployment contexts.

VI. CONCLUSION AND FUTURE WORK

A CNN-based non-invasive fingerprint blood group identification method was described in this research. Four CNN architectures were trained using a uniform preprocessing pipeline consisting of ridge segmentation, histogram equalization, and binarization.
While VGG16 attained the highest training accuracy (98.0%), ResNet showed the best generalization (81.42% validation accuracy). For well-represented blood groups, real-time forecasts produced confidence levels more than 99.8%, indicating potential for practical deployment.
Future work will focus on three directions. In order to improve model generalizability, the dataset should first be expanded through multi-institution collaborations to cover a variety of demographics. Second, investigating transfer learning to improve classification accuracy and decrease overfitting using domain-adapted pretraining and sophisticated architectures like EfficientNet or Vision Transformers.
Third, the model will be integrated into a web-based or mobile application supporting real-time fingerprint acquisition for deployment in remote and emergency healthcare environments. In high-stakes forensic situations, multimodal fusion—combining fingerprint with palm print or iris data—may further increase robustness.

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